

Image Super-Resolution Via SRCNN

Project Summary Report

1. Introduction and Problem Statement

Single Image Super-Resolution (SISR) aims to recover a high-resolution (HR) image from a single low-resolution (LR) input. This is a highly ill-posed problem since multiple HR images can correspond to a single LR image. The paper *"Image Super-Resolution Via a Convolutional Neural Network"* (SRCNN) revolutionized this domain by introducing an end-to-end deep learning framework, replacing traditional sparse-coding methods. The objective of this project is to implement SRCNN, construct a complete pipeline, evaluate its performance against interpolation baselines, and perform architectural ablation studies.

2. Dataset and Preprocessing

The dataset utilized is a 100-image subset of the DIV2K dataset, consisting of high-quality (2K resolution) images widely used in super-resolution tasks.

- **Splits:** 60 images for training, 20 for validation, and 20 for testing.
- **Pipeline:**
 1. **Center Crop:** Original images are center-cropped to 800—800 to maintain uniform spatial dimensions.
 2. **HR Target:** Cropped images are resized to 224—224 using bicubic interpolation.
 3. **LR Input:** The 224—224 images are downsampled to 112—112 (via bilinear interpolation) and then upsampled back to 224—224 (bicubic/bilinear) to match the HR spatial dimensions.
 4. **Normalization:** Pixel intensities are scaled to $[0.0, 1.0]$.

3. Model Architecture (SRCNN)

The SRCNN architecture consists of three convolutional layers designed to mimic the sparse-coding pipeline:

1. **Patch Extraction and Representation:** `Conv2d(in=3, out=64, kernel=9, pad=4) + ReLU` extracts overlapping patches and represents them as high-dimensional vectors.
2. **Non-linear Mapping:** `Conv2d(in=64, out=32, kernel=1, pad=0) + ReLU` non-linearly maps these vectors to high-resolution patch representations.
3. **Reconstruction:** `Conv2d(in=32, out=3, kernel=5, pad=2)` aggregates these representations to output the final HR image.

The output is evaluated using the **Mean Squared Error (MSE)** loss, optimizing the network to predict pixel-perfect reconstructions.

4. Experimental Setup and Baselines

To evaluate the network, we compute the **Peak Signal-to-Noise Ratio (PSNR)**, which

measures reconstruction quality logarithmically: $PSNR = 10 * \log_{10}(1.0 / MSE)$

Baselines: We establish performance benchmarks using standard interpolation methods:

- Nearest Neighbor Upsampling
- Bilinear Interpolation
- Bicubic Interpolation

Ablation Studies: The project framework supports modular testing of architectural variants:

- **BatchNorm (+ BatchNorm):** Inserting Batch Normalization after convolutions (before ReLU) to stabilize training.
- **Tanh Activation (+ Tanh):** Applying a Tanh function at the output layer to bound pixel predictions.
- **L2 Regularization (+ Regularization):** Applying weight decay ($1e-2$ to $1e-6$) to investigate its effect on overfitting on the small 60-image training set.

5. Results and Conclusion

The deep learning approach (SRCNN) is theoretically positioned to outperform conventional bicubic interpolation, especially in restoring sharp edges and high-frequency textures. During our test evaluation on 20 held-out images, the models achieved the following Peak Signal-to-Noise Ratio (PSNR):

- **Nearest Neighbor:** 23.71 dB
- **Bilinear:** 23.69 dB
- **Bicubic:** 24.96 dB
- **SRCNN (Ours):** 26.60 dB

Our trained SRCNN model successfully outperformed all standard interpolation baselines, achieving a massive **1.64 dB improvement** over the standard Bicubic upsampling algorithm.

The learning rate of 0.001 combined with data augmentation (random flips) and 200 epochs of training provided optimal convergence. Ablation studies (like adding BatchNorm) can occasionally hinder pixel-wise regression tasks by shifting statistics, whereas data augmentation helps in preventing overfitting on the highly constrained 60-image training split. Ultimately, this project successfully replicates the foundational deep learning SISR technique, providing a comprehensive toolkit for training, evaluation, and interactive demonstration.